Palo Alto Networks PAN-OS — Complete Upgrade Runbook

Comprehensive Enterprise Standard Operating Procedure: Pre-Upgrade Readiness, Sequential Major Version Pathways, Web GUI & CLI Execution, HA Zero-Downtime Failover, and Disaster Rollback

Document ID:	SOP-PAN-UPG-01	**Hardware & VM:**	PA-220 / 400 / 800 / 3200 / 5200 / 7000 / PA-VM
Release Trains:	PAN-OS 9.1 → 10.0 → 10.1 → 10.2 → 11.0 → 11.1	**HA Deployment:**	Active/Passive & Standalone Appliances
Classification:	Critical Infrastructure Maintenance	**Lead Engineer:**	Nishu Saini (Innspark Solutions)

1. Fundamental Architectural Rules & Version Hop Constraints

Upgrading Palo Alto PAN-OS requires adherence to strict architectural rules. Violating upgrade rules can result in XML schema corruption, failed configuration commits, daemon crashes, or unbootable storage partitions. The following core principles govern every upgrade:

GOLDEN RULES OF PAN-OS UPGRADES

1. Sequential Upgrade Path (NO Major Version Skipping): You cannot hop directly across major versions (e.g. 9.1.x directly to 10.2.x is prohibited). You must traverse every intermediate major release: **9.1.x** → **10.0.x** → **10.1.x** → **10.2.x**.
2. Base (.0) Image Requirement: To install any maintenance release (e.g., 10.1.10), the major release **base image (10.1.0) must be downloaded** onto local flash storage. However, *you do not need to install or reboot into the .0 base image*; download the .0 base image, then download and install your target maintenance release directly.
3. Dynamic Updates Prerequisite: The Applications and Threats database must be updated to the latest available package before triggering any PAN-OS software upgrade.
4. Panorama Central Management Rule: If managed centrally via Panorama, Panorama must ALWAYS be upgraded to a version equal to or higher than the firewalls it manages.

Starting Release	Target Release	Mandatory Step-by-Step Upgrade Sequence
`PAN-OS 9.1.x`	**PAN-OS 10.1.x (Recommended)**	1. Download 10.0.0 (Base) → Download & Install 10.0.x → Reboot. 2. Download 10.1.0 (Base) → Download & Install 10.1.10 → Reboot.
`PAN-OS 10.1.x`	**PAN-OS 10.2.x**	1. Download 10.2.0 (Base) → Download & Install target 10.2.x (e.g. 10.2.8-h3) → Reboot.
`PAN-OS 10.2.x`	**PAN-OS 11.0.x / 11.1.x**	1. Download 11.0.0 (Base) → Install 11.0.x → Reboot → Download 11.1.0 → Install 11.1.x → Reboot.

2. Pre-Upgrade Preparation & Readiness Checklist

Execute these pre-flight checks before scheduling the maintenance window:

Check Item	Action / CLI Verification Command	Success Criterion
1. Full Off-Box Backup	save config to Pre_Upgrade_Backup.xml Export named configuration snapshot via Web GUI to external drive.	Verified XML backup file stored outside the firewall.
2. Disk Space Audit	show system disk-space Check /opt/pancfg and / partitions.	Partition utilization must be below 85%. Delete stale software images if needed.
3. Clean Old Software	delete software version <old-version> Purge outdated versions from flash storage.	At least 3.0 GB of free space available on root/panos partition.
4. Dynamic Updates	request content upgrade check Install latest *Applications and Threats* package.	App/Threat engine version current and status = Completed.
5. Candidate Config	show config diff	No uncommitted changes in candidate buffer prior to upgrade.
6. Active Sessions & HA	show session info show high-availability state	Firewall healthy; HA synchronization state = **synchronized**.

3. Web GUI Step-by-Step Upgrade Procedure

STEP 1: Update Content Engine (Applications & Threats)

1. In the Web GUI, navigate to: **Device** → **Dynamic Updates**.
2. Click **Check Now** at the bottom of the screen.
3. Locate the latest version of **Applications and Threats**.
4. Click **Download**, and once finished, click **Install**.
5. Verify task completion in the **Tasks** window (Status = Completed).

STEP 2: Download Mandatory Base (.0) Image

1. In the Web GUI, navigate to: **Device** → **Software**.
2. Click **Check Now** at the bottom of the screen to poll Palo Alto cloud update servers.
3. Locate the **Base Image (.0)** of the next major version (e.g., PanOS_10.1.0).
4. In the *Action* column, click **Download**.
5. *Crucial:* Once the download finishes, **DO NOT CLICK INSTALL** on the .0 base image.

[OPERATIONAL NOTE]: Base images must only be downloaded, not installed.

STEP 3: Download and Install Target Maintenance Release

1. In the same **Device** → **Software** window, locate the target maintenance release (e.g., PanOS_10.1.10).
2. Click **Download**. Monitor the task progress via the bottom-right **Tasks** window.
3. Once downloaded, click **Install** in the Action column.
4. An installation pop-up will appear asking: *'Do you want to reboot the device after installation?'*
   - If inside your maintenance window: Select **Yes** to automatically reboot.
   - If staging ahead of time: Select **No** (reboot later manually via Device → Operations → Reboot).

STEP 4: Reboot and Post-Upgrade Dashboard Verification

1. During reboot, the management plane will be offline for approximately 5 to 10 minutes (hardware platforms may take 12–15 minutes).
2. Ping the management IP continuously: ping -t 172.17.0.50.
3. Once ICMP replies resume, navigate to **https://172.17.0.50** and log into the Web GUI.
4. On the **Dashboard**, inspect the **General Information** widget:
   - **Software Version:** Confirm it reflects the new target version (e.g. 10.1.10).
   - **Application Version:** Confirm current.
   - **Threat Version:** Confirm current.

4. Alternative PAN-OS CLI Commands (Direct Terminal Execution)

For network engineers performing upgrades via SSH or serial console, use this verified end-to-end command sequence:

```
# Phase 1: Pre-Upgrade Verification & Content Update
admin@PA-1> show system info
admin@PA-1> show system disk-space
admin@PA-1> request content upgrade check
admin@PA-1> request content upgrade download latest
admin@PA-1> request content upgrade install version latest

# Phase 2: Query Cloud Servers & Download Base Software Image
admin@PA-1> request system software check
admin@PA-1> request system software download version 10.1.0

# Phase 3: Download and Install Target Maintenance Release
admin@PA-1> request system software download version 10.1.10
admin@PA-1> request system software install version 10.1.10

# Phase 4: Monitor Software Installation Job
admin@PA-1> show jobs all
# (Look for job type: 'SWInstall', Status: 'FIN', Result: 'OK')

# Phase 5: Initiate Graceful System Reboot
admin@PA-1> request restart system
# Press 'y' to confirm system restart

# Phase 6: Post-Reboot Verification & Validation
admin@PA-1> show system info | match sw-version
# (Expected: sw-version: 10.1.10)
admin@PA-1> show routing route summary
admin@PA-1> show session info
```

5. Air-Gapped / Offline Upgrade Method (No Internet Access)

For isolated defense, financial, or SCADA environments where the firewall MGT interface cannot access the public internet:

OFFLINE / AIR-GAPPED DEPLOYMENT RUNBOOK

1. **Download from Support Portal:** On an internet-connected workstation, log into support.paloaltonetworks.com → *Updates* → *Software Updates*. Download both the Base image (.0) and target maintenance release (.xfr / .panos).
2. **Upload via Web GUI:** Navigate to **Device** → **Software** → **Upload**. Click Browse, select the downloaded image file, and click OK.
3. **Upload via SCP (CLI):** From terminal, SCP the software package directly into the firewall's panos repository:
   scp PanOS_10.1.10 admin@172.17.0.50:/opt/panrepo/
4. **Install via CLI:** request system software install file PanOS_10.1.10 followed by request restart system.

6. High Availability (HA Active/Passive) Zero-Downtime Upgrade Sequence

In production environments with Active/Passive HA pairs, perform the upgrade without dropping live customer traffic using this exact sequence:

Step	Target Appliance	Action Performed	Traffic Impact
1. Pre-Check	Both Nodes (FW-A & FW-B)	Verify HA sync: show high-availability state. Ensure sync = synchronized.	None (FW-A active).
2. Upgrade Passive	**Passive Peer (FW-B)**	Download base and target images on FW-B. Install target version and reboot FW-B.	None (FW-A handles 100% traffic).
3. Verify Passive	Passive Peer (FW-B)	FW-B boots into new PAN-OS. Verify status = **Passive** and HA sync = **synchronized**.	None (FW-A handles 100% traffic).
4. Controlled Failover	Active Peer (FW-A)	Suspend FW-A to gracefully shift traffic: request high-availability state suspend.	Sub-second failover to FW-B.
5. Upgrade Former Active	**Suspended Peer (FW-A)**	Download base and target images on FW-A. Install target version and reboot FW-A.	None (FW-B handles 100% traffic).
6. Restore Normal State	Former Active (FW-A)	Once FW-A boots: request high-availability state functional. Both peers now run identical new PAN-OS.	Full HA redundancy restored.

7. Disaster Recovery: Emergency Rollback to Previous PAN-OS Release

Palo Alto firewalls employ a dual-boot partition architecture (sysroot0 and sysroot1). When you upgrade, the new OS is written to the inactive partition, leaving the previous working operating system intact on the alternate partition. If severe unexpected packet drops or daemon instability occur post-upgrade:

```
# Method A: Software Revert via Running CLI (Within Current Session)
admin@PA-1> request system software revert
# System prompts: 'Are you sure you want to revert to the previous software version? (y or n)'
# Press 'y' -> PAN-OS switches the bootloader pointer to the previous partition.
admin@PA-1> request restart system

# Method B: Rollback via Maintenance Mode (maint) Console
# 1. Connect serial console (9600-8-N-1) -> Reboot appliance.
# 2. Type 'maint' at boot countdown -> Select 'Continue'.
# 3. Navigate to: 'Select Root Partition' (or 'Boot Inactive Partition').
# 4. Choose the previous working sysroot partition (shows previous PAN-OS version) -> Select 'Reboot'.
```

OPERATIONAL SIGNOFF & VALIDATION

Post-Upgrade Maintenance Window Closure: Monitor firewall CPU utilization (show running resource-monitor) and data plane packet buffers for 60 minutes post-upgrade. Verify all IPSec VPN tunnels (show vpn ike-sa, show vpn ipsec-sa) and external BGP/OSPF peerings are fully established before declaring maintenance completion.